

Espresso Machine

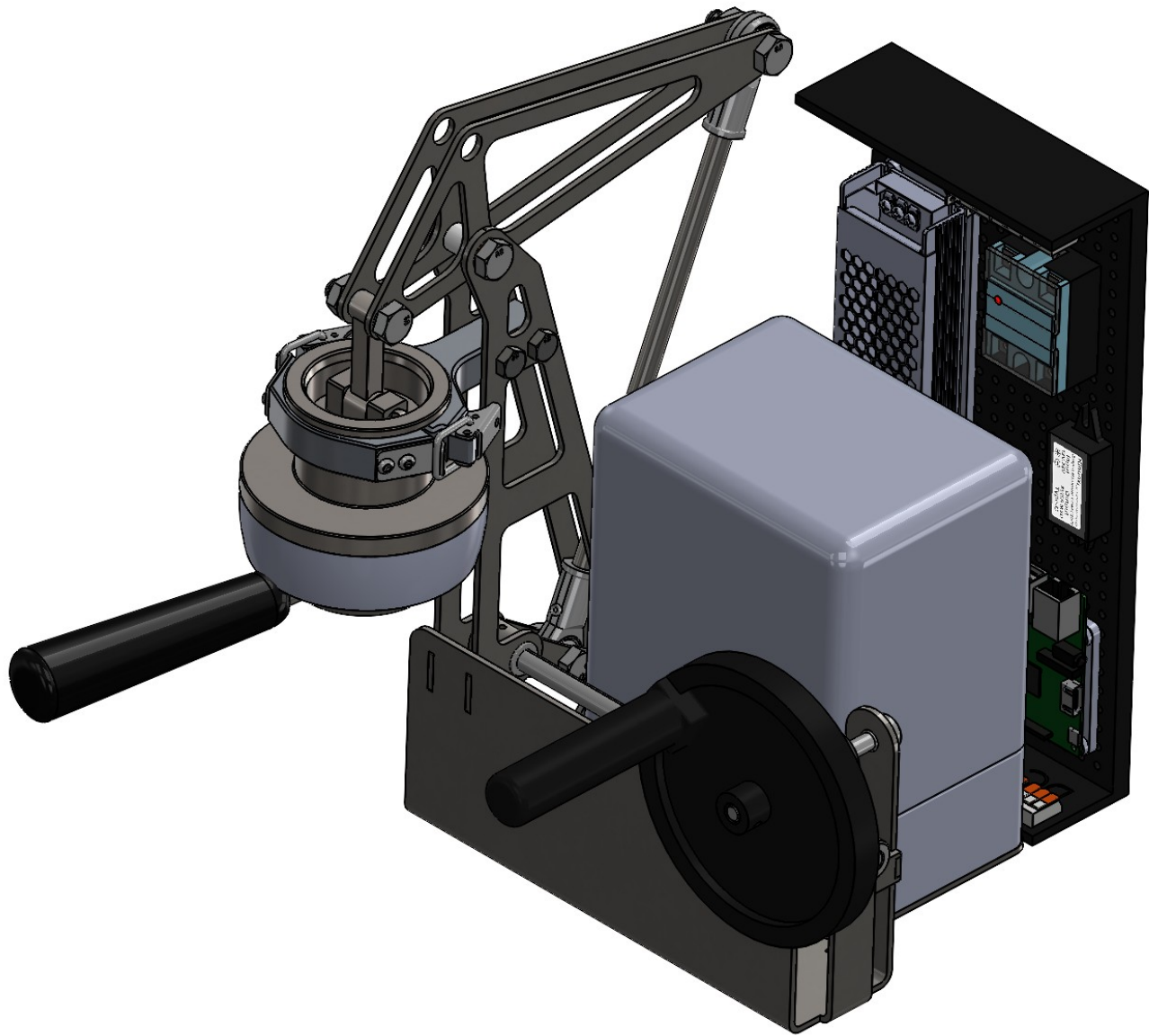
Westley Baird, Alexis Phipps, Josua Albertsson, Gabriel Camacho, Joey Dwyer

Group 1 – ME 190 Spring 2026

Drawing Package



SACRAMENTO
STATE



DEFAULT
TOLERANCES:
ANGLE ± 0.5°
X ± 2.5
.X ± 0.25
.XX ± .13
.XXX ± .013

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Full Assembly

PART/DWG #: 1

REV: 5

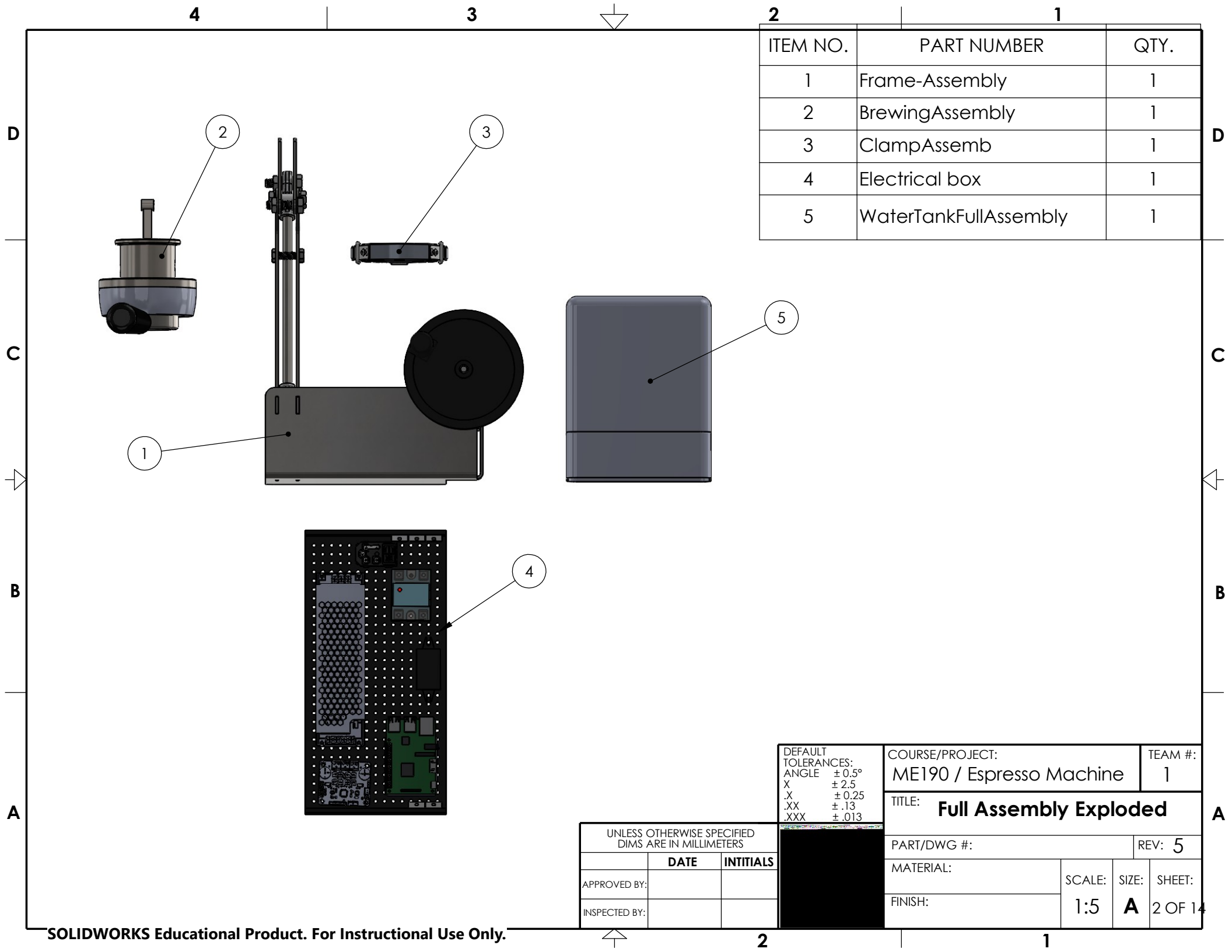
MATERIAL:

SCALE: SIZE: SHEET:

FINISH:

1:3 A 1 OF 1

UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



ITEM NO.	PART NUMBER	QTY.
1	Frame-Assembly	1
2	BrewingAssembly	1
3	ClampAssemb	1
4	Electrical box	1
5	WaterTankFullAssembly	1

DEFAULT
TOLERANCES:
ANGLE ± 0.5°
X ± 2.5
.X ± 0.25
.XX ± .13
.XXX ± .013

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Full Assembly Exploded

PART/DWG #:

REV: 5

MATERIAL:

SCALE: SIZE: SHEET:

FINISH:

1:5 A 2 OF 14

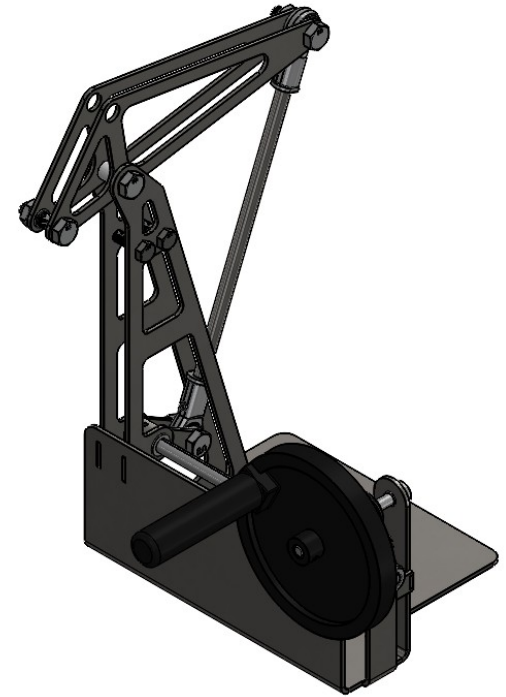
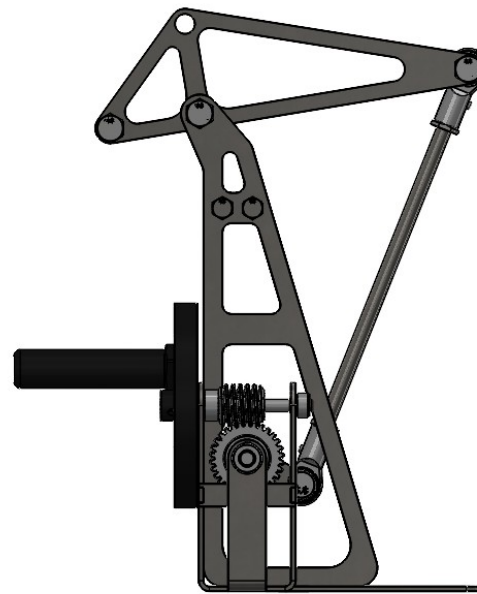
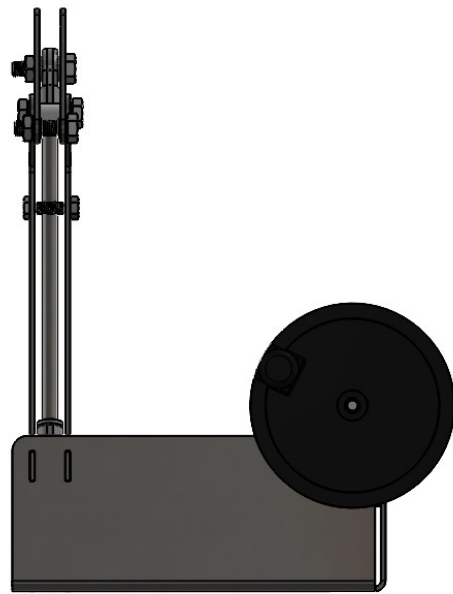
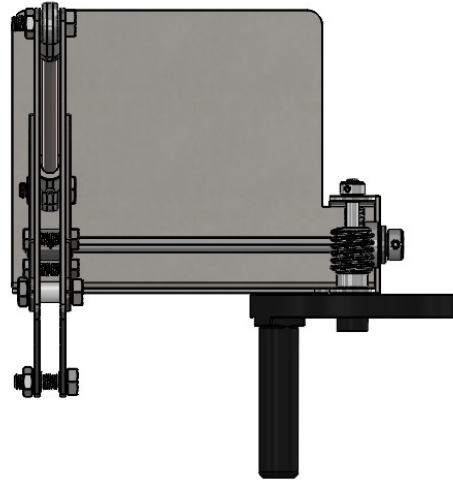
UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		

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2

1



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
.X $\pm .1$
.XX $\pm .03$
.XXX $\pm .005$

COURSE/PROJECT:
ME 190 / Espresso Machine

TEAM #:
1

TITLE: **Frame Assembly**

PART/DWG #: REV: 1

MATERIAL: SCALE: SIZE: SHEET: 14 OF 4

FINISH: 1:16 A

UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



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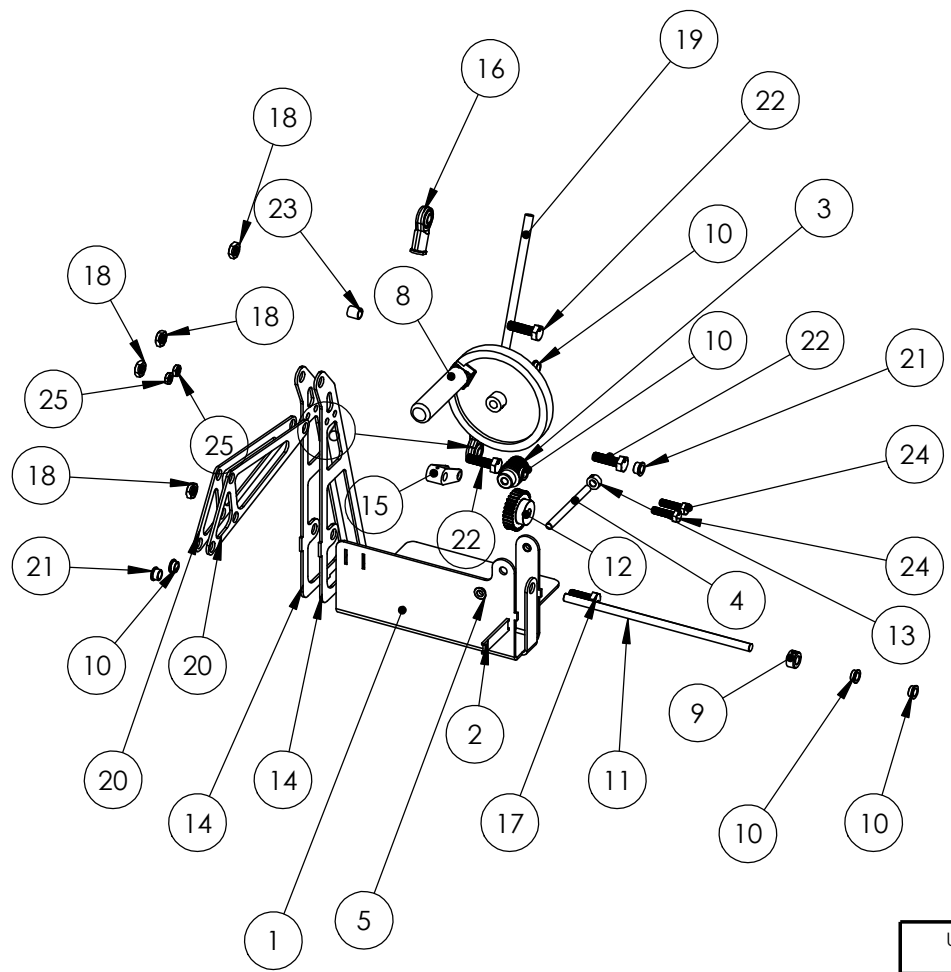
2

1

Purchased: 3,4,5,9,10,11,12,13,16,17,18,19,21,22,24,25

Indexing Missing 6-7 due to BOM import discrepancy

ITEM NO.	PART NUMBER	Material	QTY.
1	Baseplate	AISI 304 Stainless Steel	1
2	Bracket Support	AISI 304 Stainless Steel	1
3	57545K631_Steel Worm	1045 Carbon Steel	1
4	4143N17	303 Stainless Steel	1
5	6659K224	841 Bearing Bronze	2
8	spiillllllnNNNnnN	PolyPropylene	1
9	6058N18	303 Stainless Steel	1
10	6659K146	841 Bearing Bronze	5
11	8632T38	Carbon Steel	1
12	57545K758_Metal Worm Gear	Bronze	1
13	6056N16	Carbon Steel	1
14	UrRightv2	AISI 304 Stainless Steel	2
15	LinkageArm	AISI 304 Stainless Steel	1
16	6072K531	Carbon Steel	2
17	91180A634	Plain Carbon Steel	1
18	90364A102	Steel	4
19	M10 x 1.25 x 250mm AllThreadRod	AISI 304 Stainless Steel	1
20	LeverArm	AISI 304 Stainless	2
21	6659K117	841 Bearing Bronze	2
22	91280A371	Plain Carbon Steel	3
23	3dPrinted Spacer	ABS	1
24	91280A085	Plain Carbon Steel	2
25	94168A114	Steel	2



UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		

DEFAULT TOLERANCES:
ANGLE ± 0.5°
.X ± .1
.XX ± .03
.XXX ± .005


SACRAMENTO STATE

COURSE/PROJECT: ME 190 / Espresso Machine		TEAM #: 1	
TITLE: Frame Assembly			
PART/DWG #:		REV: 1	
MATERIAL:	SCALE: 1:20	SIZE: A	SHEET: 3 OF 14
FINISH:			

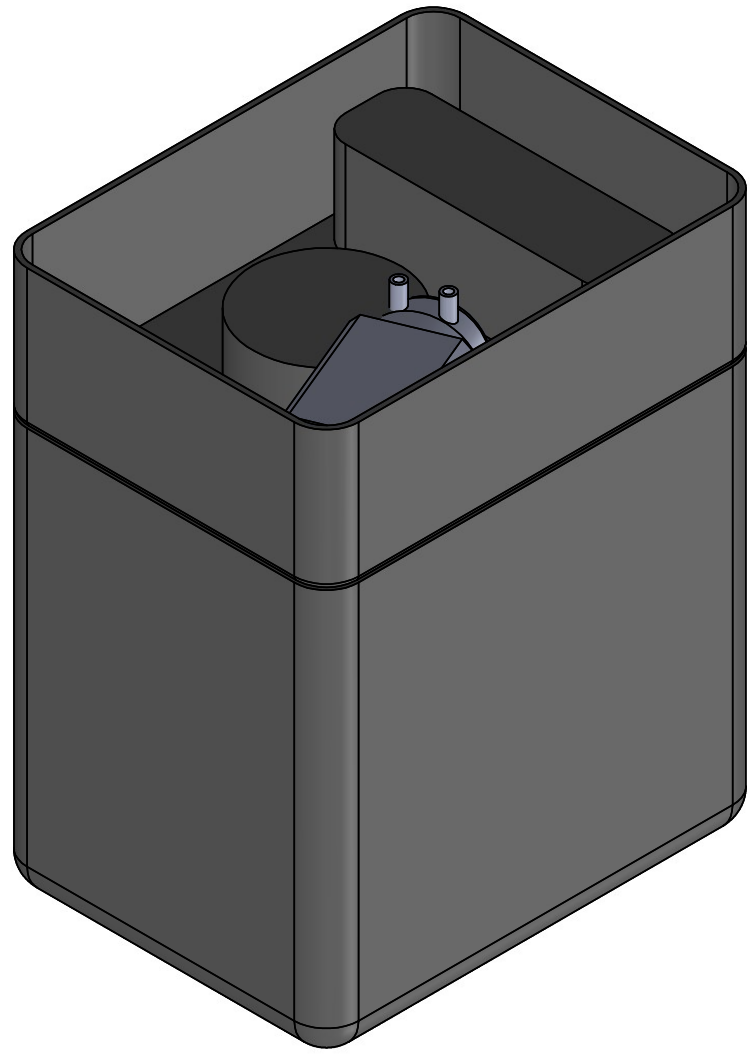
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REVISIONS			
REV.	DRAWN BY JOSUA ALBERTSSON		DATE
			APPROVED



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
X ± 2.5
.X $\pm .25$
.XX $\pm .13$

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE: **Water Tank Assembly
ISOMETRIC**

PART/DWG #: N/A

REV: 1

MATERIAL:
PEHD/302SS/Silicon

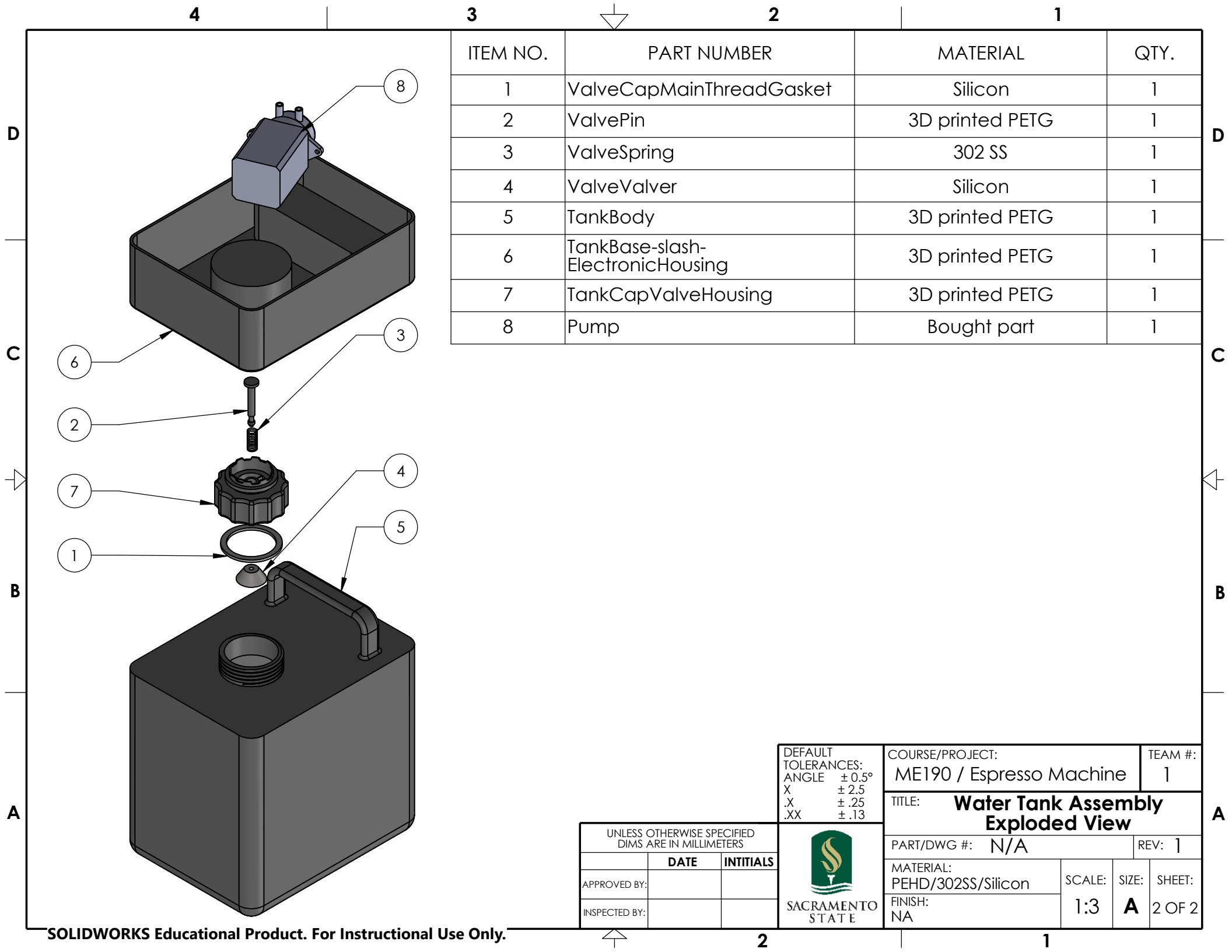
SCALE: SIZE: SHEET:

FINISH:
NA

1:2 A 1 OF 2

UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		





ITEM NO.	PART NUMBER	MATERIAL	QTY.
1	ValveCapMainThreadGasket	Silicon	1
2	ValvePin	3D printed PETG	1
3	ValveSpring	302 SS	1
4	ValveValver	Silicon	1
5	TankBody	3D printed PETG	1
6	TankBase-slash-ElectronicHousing	3D printed PETG	1
7	TankCapValveHousing	3D printed PETG	1
8	Pump	Bought part	1

DEFAULT TOLERANCES: ANGLE ± 0.5° X ± 2.5 .X ± .25 .XX ± .13	COURSE/PROJECT: ME190 / Espresso Machine			TEAM #: 1	
	TITLE: Water Tank Assembly Exploded View				
	PART/DWG #: N/A			REV: 1	
	MATERIAL: PEHD/302SS/Silicon		SCALE: 1:3	SIZE: A	SHEET: 2 OF 2
FINISH: NA					



SACRAMENTO STATE

UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



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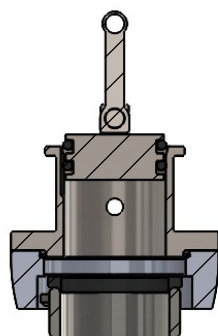
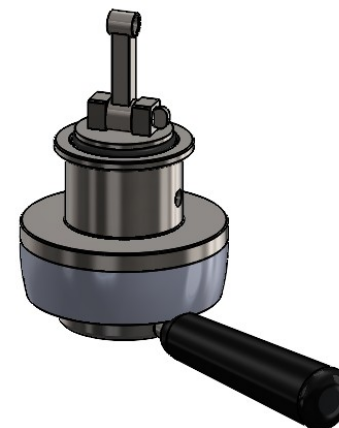
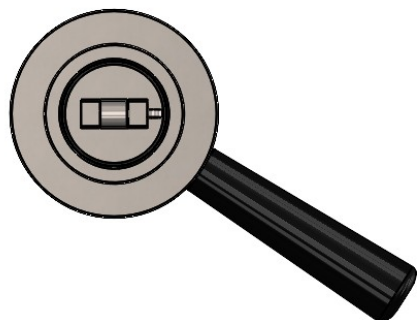
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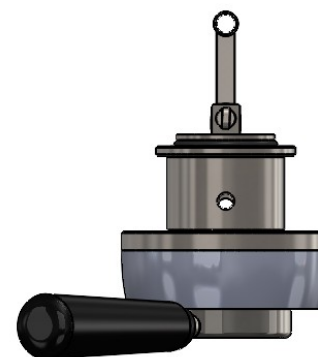
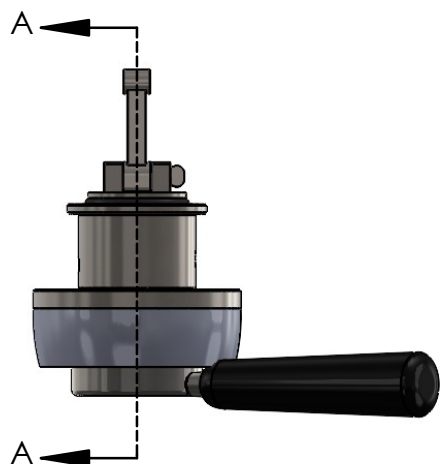
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SECTION A-A



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
X ± 2.5
.X $\pm .25$
.XX $\pm .13$
.XXX $\pm .013$

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Brew Assembly

PART/DWG #:

REV: 2

MATERIAL:

SCALE: SIZE: SHEET:

FINISH:

1:4 **A** 5 OF 14

UNLESS OTHERWISE SPECIFIED
DIMS ARE IN MILLIMETERS

DATE

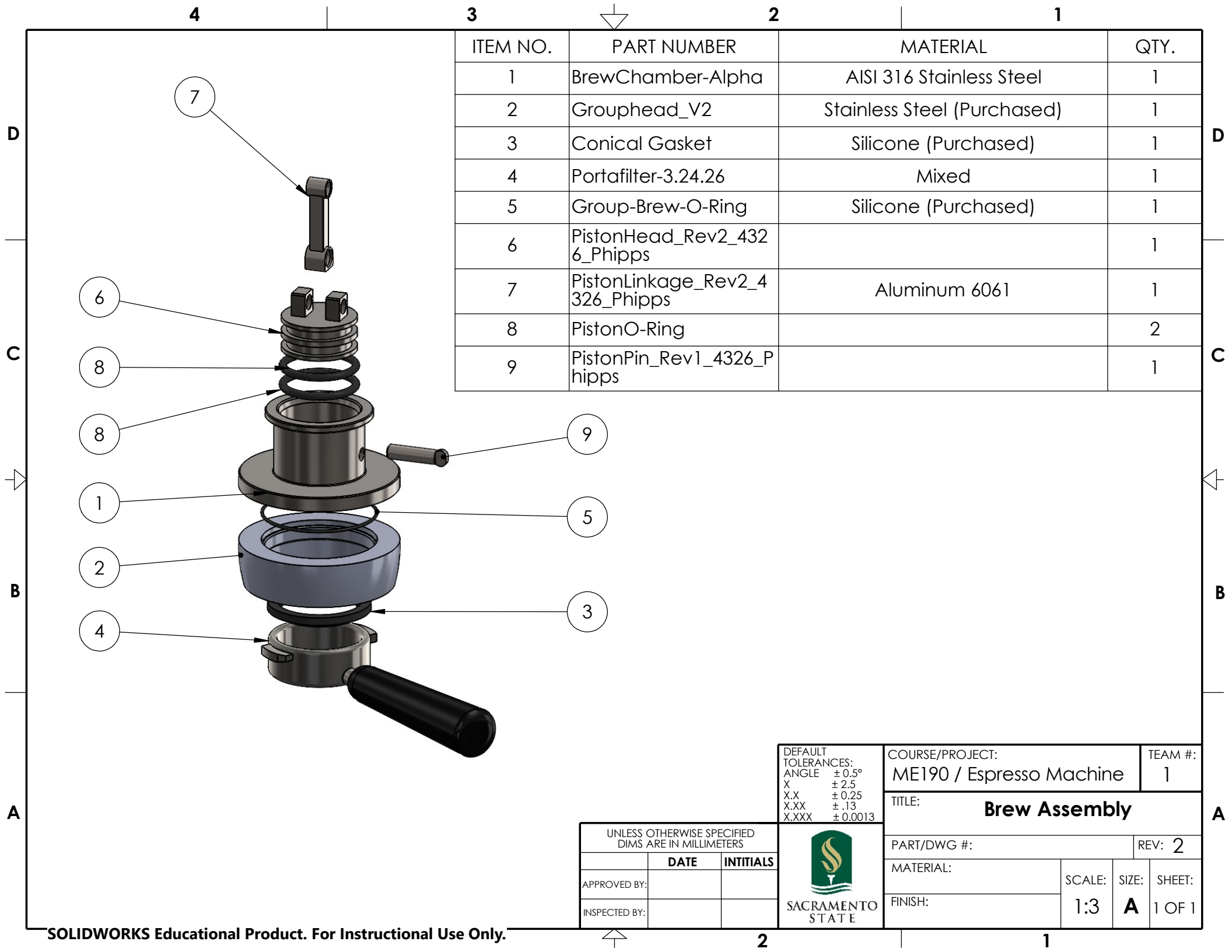
INITIALS

APPROVED BY:

INSPECTED BY:



SACRAMENTO
STATE



ITEM NO.	PART NUMBER	MATERIAL	QTY.
1	BrewChamber-Alpha	AISI 316 Stainless Steel	1
2	Grouphead_V2	Stainless Steel (Purchased)	1
3	Conical Gasket	Silicone (Purchased)	1
4	Portafilter-3.24.26	Mixed	1
5	Group-Brew-O-Ring	Silicone (Purchased)	1
6	PistonHead_Rev2_4326_Phipps	Aluminum 6061	1
7	PistonLinkage_Rev2_4326_Phipps		1
8	PistonO-Ring		2
9	PistonPin_Rev1_4326_Phipps		1

DEFAULT TOLERANCES:
ANGLE ± 0.5°
X ± 2.5
X.X ± 0.25
X.XX ± .13
X.XXX ± 0.0013

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Brew Assembly

PART/DWG #:

REV: 2

MATERIAL:	SCALE:	SIZE:	SHEET:
FINISH:	1:3	A	1 OF 1

UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



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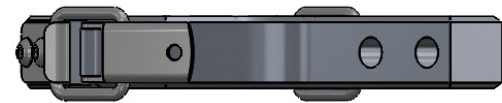
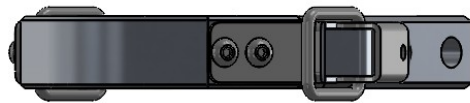
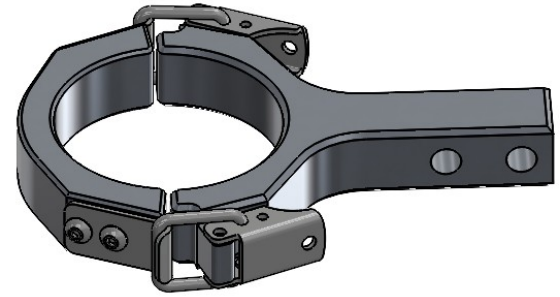
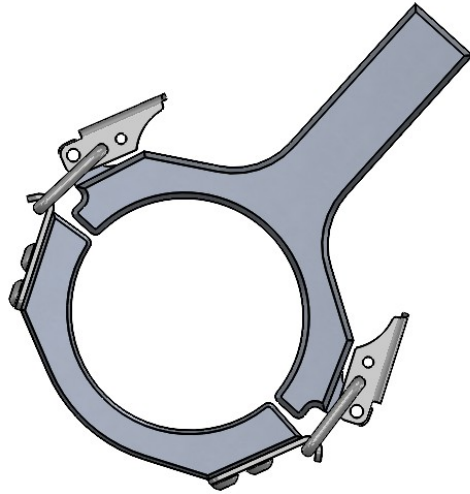
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B

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A



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
X ± 2.5
.X $\pm .25$
.XX $\pm .13$
.XXX $\pm .013$

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Brew Chamber Clamp

PART/DWG #:

REV: 3

MATERIAL:
Mixed

SCALE:
1:2

SIZE:
A

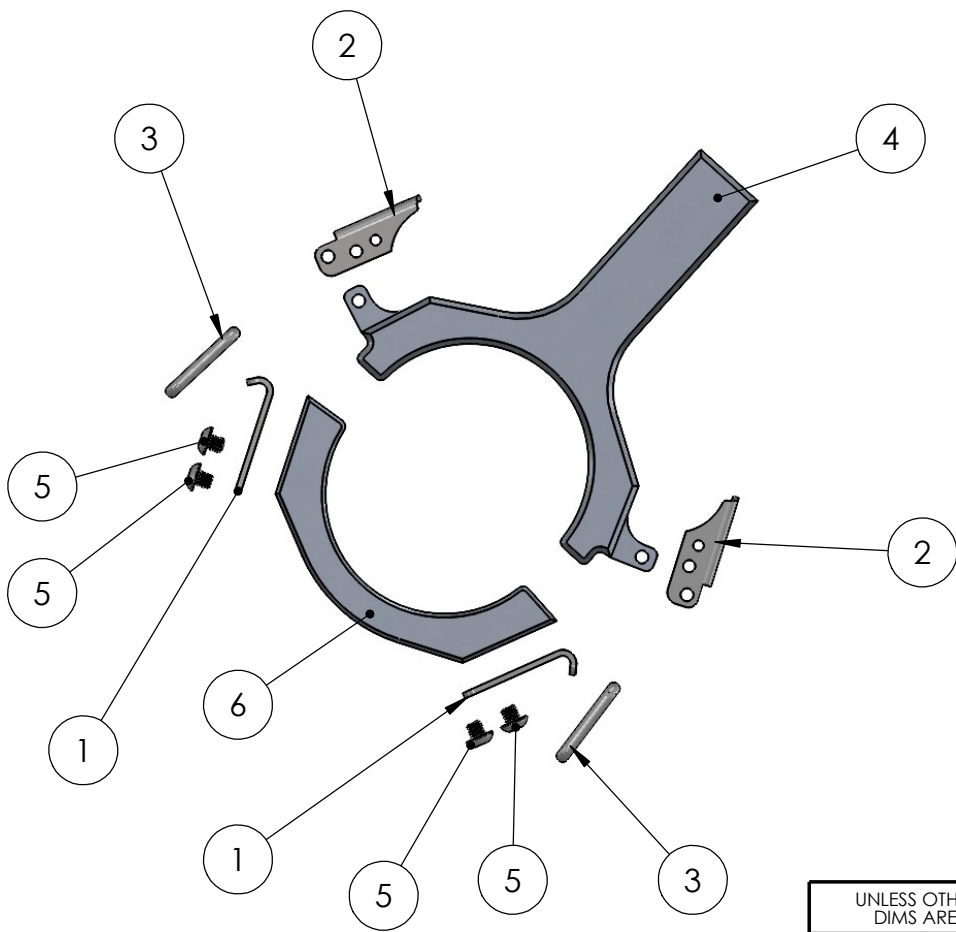
SHEET:
7 OF 14

FINISH:

UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



ITEM NO.	PART NUMBER	MATERIAL	QTY.
1	6082A16 Hook	304 Stainless Steel	2
2	6082A16 Handle	304 Stainless Steel	2
3	6082A16 Latch Wire	304 Stainless Steel	2
4	Clamp	Aluminum 6061 - T6	1
5	92095A476 Button Hex Screws	Stainless Steel (Purchased)	4
6	ClampOuter	Aluminum 6061 T6	1



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
X ± 2.5
.X $\pm .25$
.XX $\pm .13$
.XXX $\pm .013$

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE:
Brew Chamber Clamp

PART/DWG #: REV: 3

MATERIAL:
Mixed
FINISH:
SCALE: 1:2
SIZE: A
SHEET: 1 OF 1

UNLESS OTHERWISE SPECIFIED DIMS ARE IN MILLIMETERS		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



4

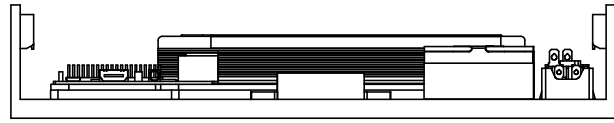
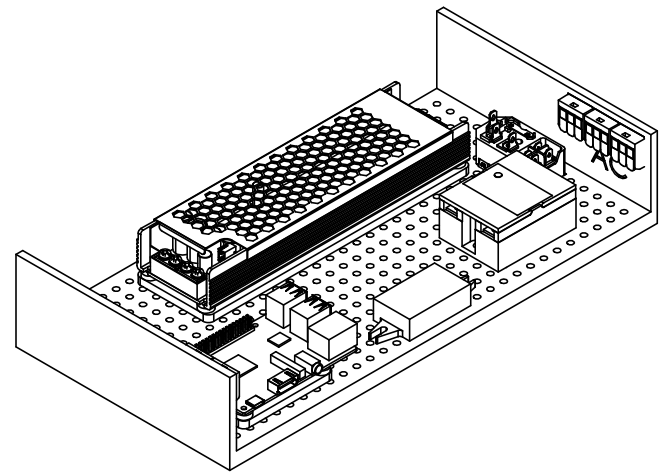
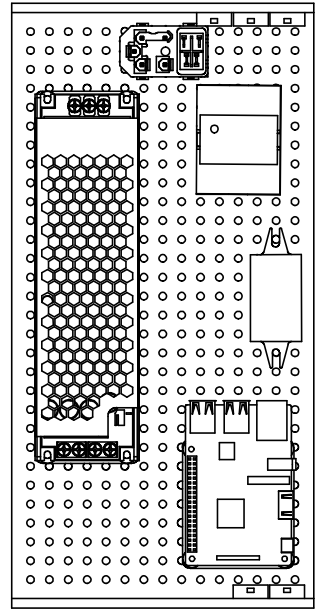
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2

1

NOTES:
1. DEBURR ALL SHARP EDGES

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	Initial Model to hold electronics	3/22/2026	Joseph
2	Shrunk model to fit in assembly	3/24/2026	Joseph



DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
X ± 2.5
.X $\pm .25$
.XX $\pm .13$

COURSE/PROJECT:
ME190 / Espresso Machine

TEAM #:
1

TITLE: **Electrical box assembly**

PART/DWG #:

REV: 2

MATERIAL:

SCALE:

SIZE:

SHEET:

FINISH:

1:4

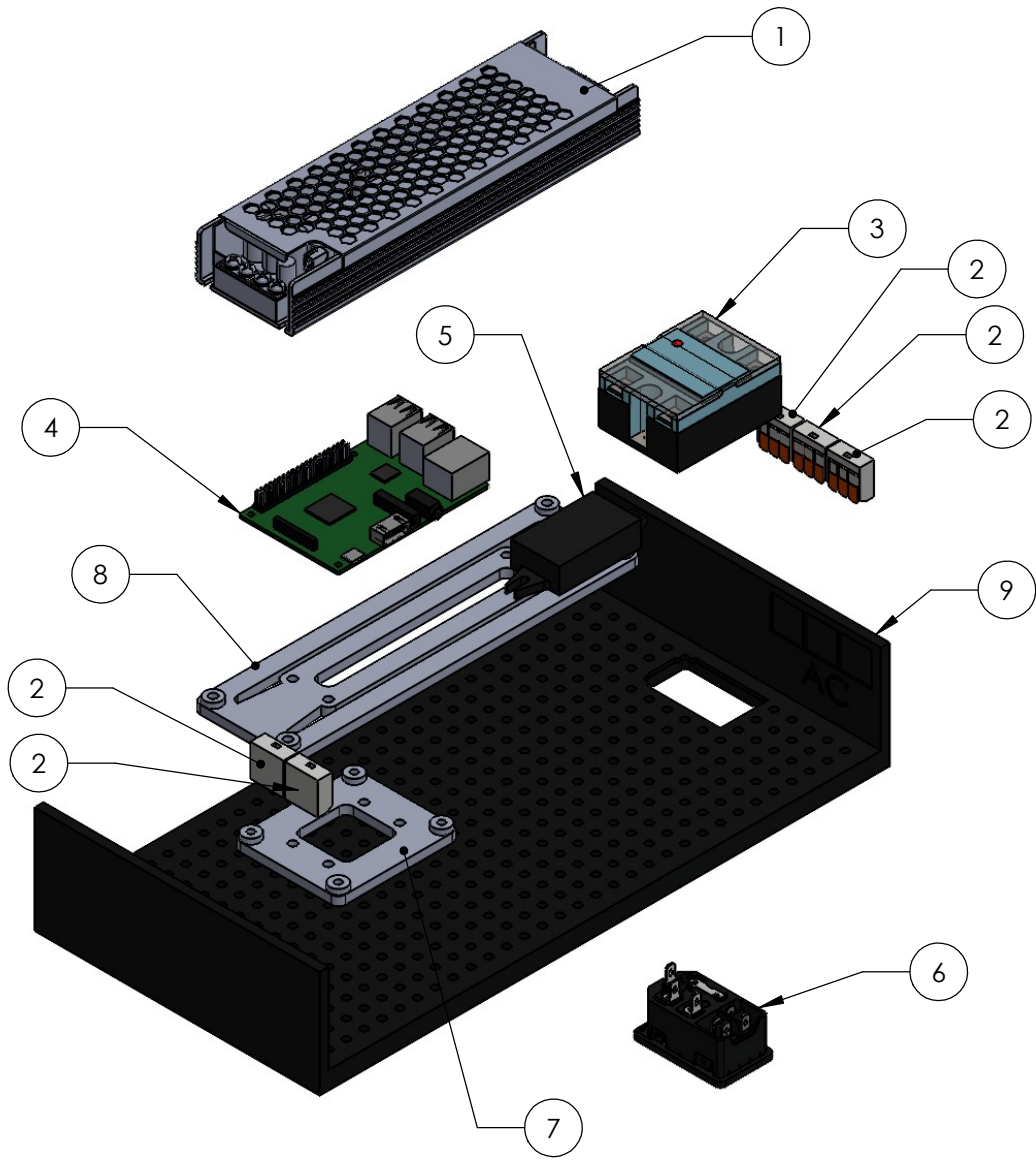
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10 OF 4

UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		

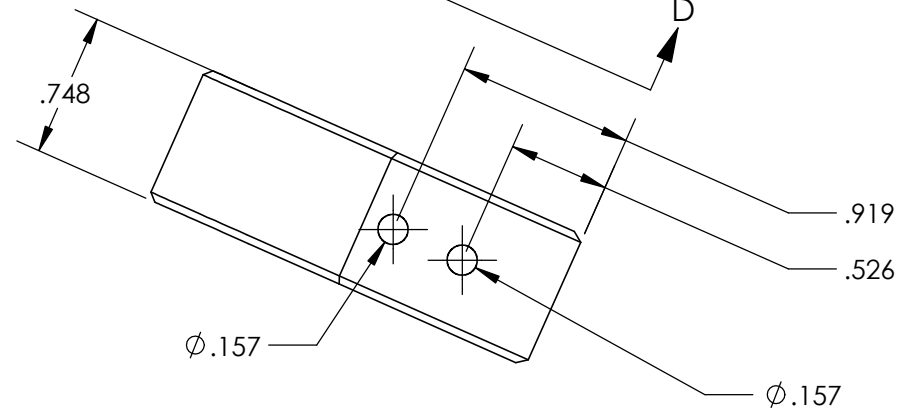
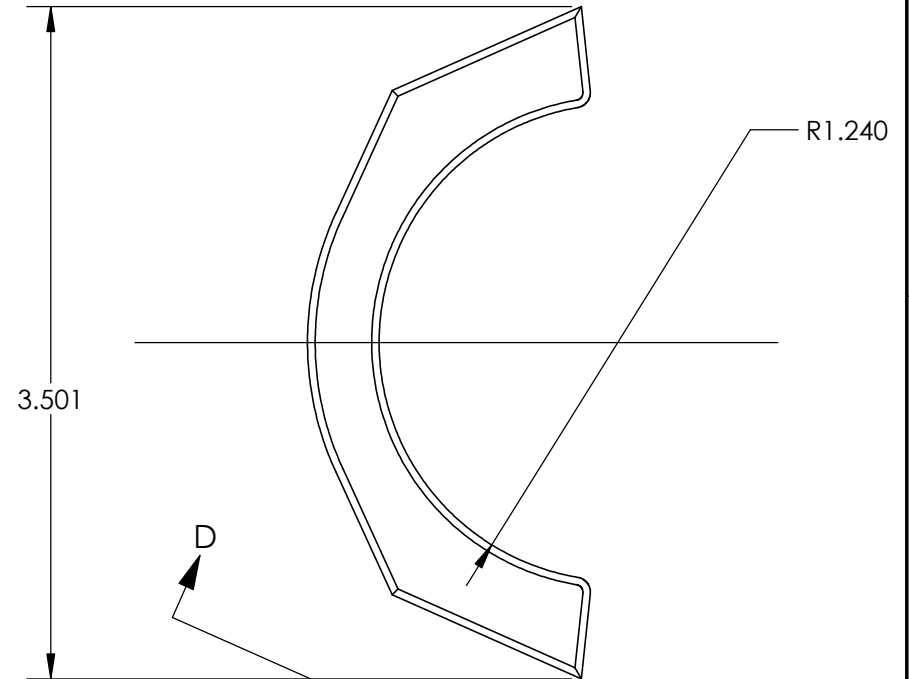
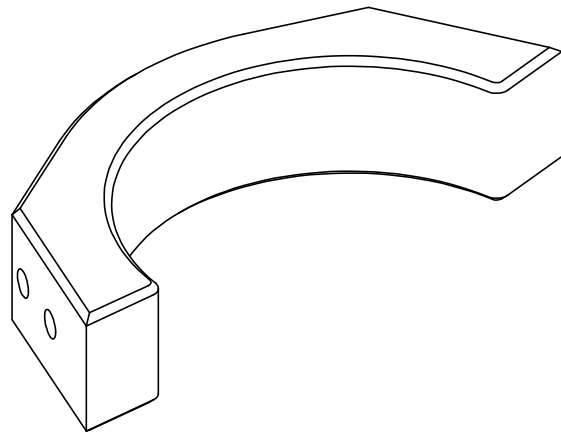


NOTES:
1. DEBURR ALL SHARP EDGES



Item No.	Part	Material	QTY
1	120V AC to 24V DC 300w PSU	Mixed	1
2	Wago	Mixed	5
3	Omoron SSR	Mixed	1
4	Raspberry Pi 3B+	Mixed	1
5	24V DC to 5V DC USB C output PSU	Mixed	1
6	Filtered power inlet	Mixed	1
7	Pi mount	ASA	1
8	PSU Mount	ASA	1
9	Enclosure Bottom	ASA	1

DEFAULT TOLERANCES: ANGLE ± 0.5° X ± 2.5 .X ± .25 .XX ± .13	COURSE/PROJECT: ME190 / Espresso Machine			TEAM #: 1	
	TITLE: Electrical box assembly				
 SACRAMENTO STATE	PART/DWG #:			REV: 2	
	MATERIAL:		SCALE: 1:3	SIZE: A	SHEET: 9 OF 14
	FINISH:				



VIEW D-D

DRAWN BY:
WESTI FY BAIRD

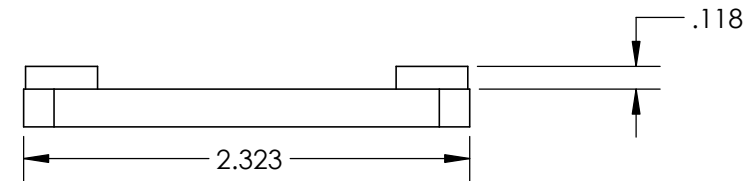
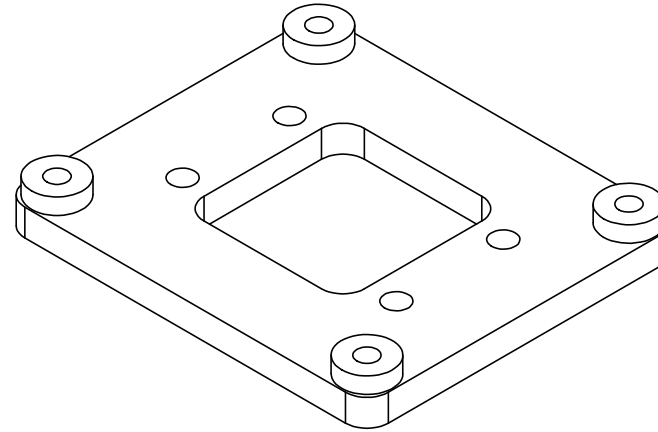
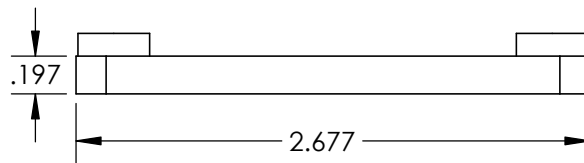
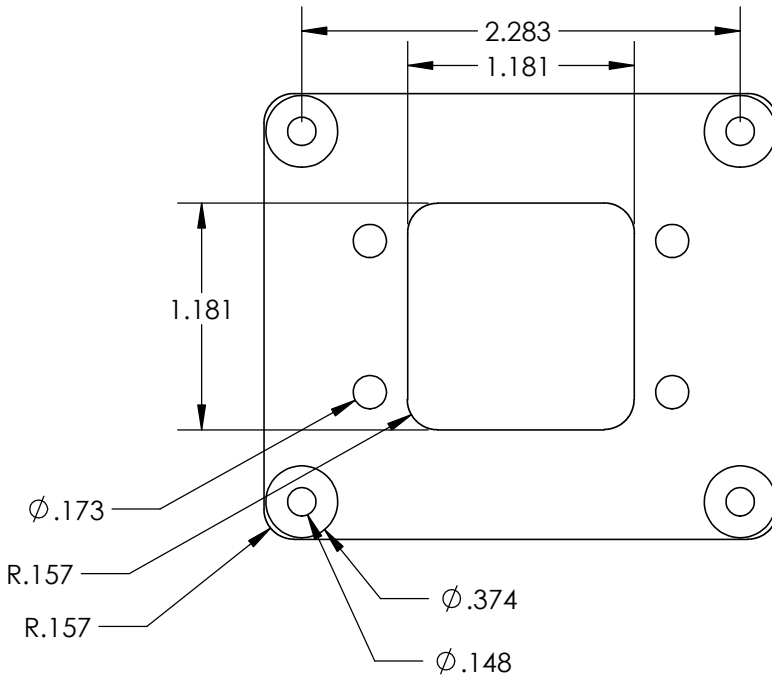
UNLESS OTHERWISE SPECIFIED DIMS ARE IN MM		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		

DEFAULT
TOLERANCES:
ANGLE $\pm 0.5^\circ$
.X $\pm .1$
.XX $\pm .03$
.XXX $\pm .005$



COURSE/PROJECT: ME 190 / ESPRESSO M.			TEAM #: 1	
TITLE: Outer Clamp				
PART/DWG #:			1	REV: 1
MATERIAL: Aluminum 6061 T6		SCALE: 1:1	SIZE: A	SHEET: 8 OF 14
FINISH: AS RECEIVED				

NOTES:
1. DEBURR ALL SHARP EDGES



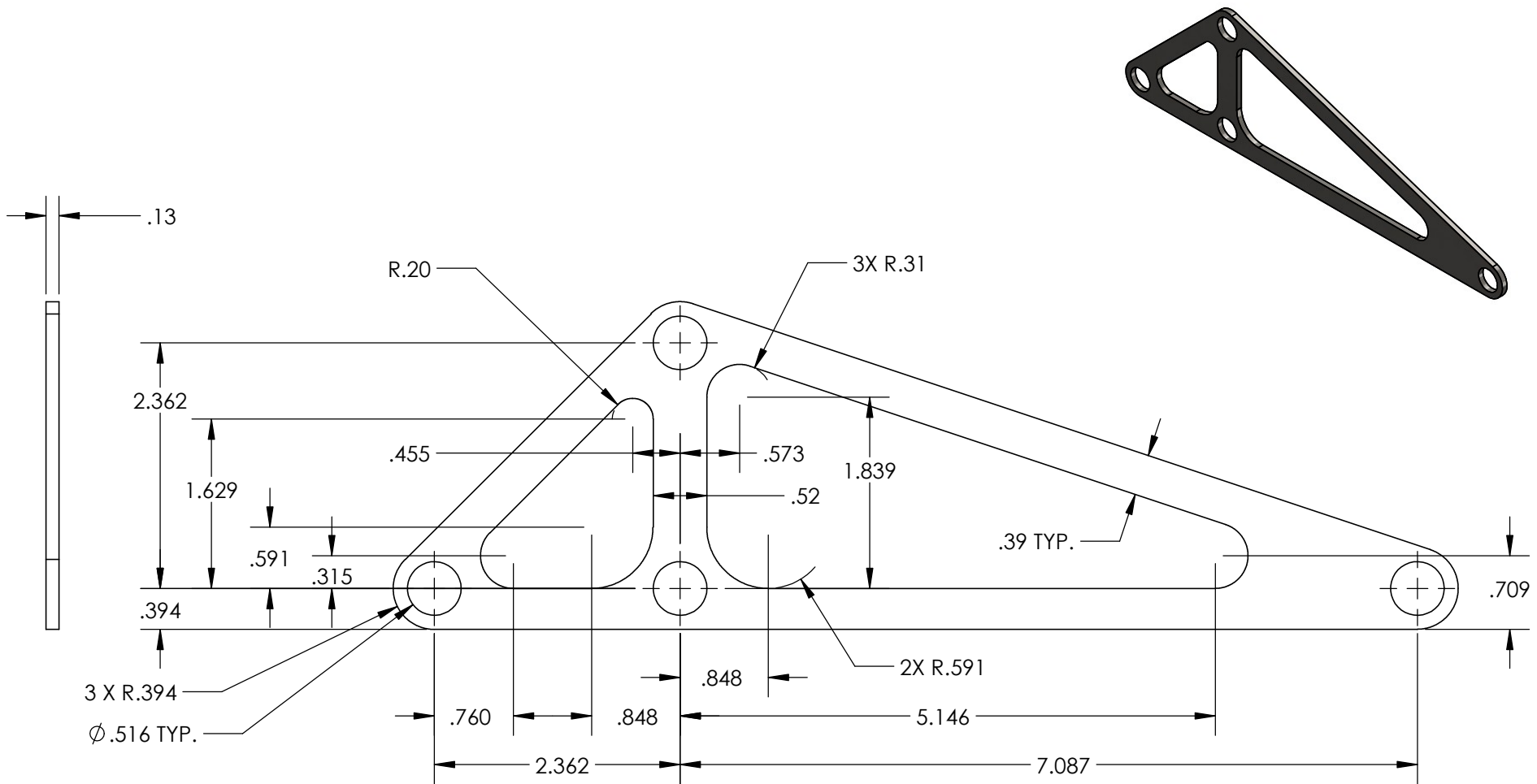
REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	Make a mount compatible between Pi and Enclosure	4/26/2026	Joseph

UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		

DEFAULT TOLERANCES: ANGLE ± 0.5° .X ± .1 .XX ± .03 .XXX ± .005	COURSE/PROJECT: ME190 / Slp Happens			TEAM #: 1	
	TITLE: Pi Mount				
 SACRAMENTO STATE	PART/DWG #: 1			REV: 1	
	MATERIAL: ASA		SCALE: 1:1	SIZE: A	SHEET: 11 OF 14
	FINISH: 3D Printed Raw				

NOTES:
1. DEBURR ALL SHARP EDGES

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED



DEFAULT TOLERANCES: ANGLE $\pm 0.5^\circ$ X ± 2.5 .X ± 0.23 .XX $\pm .13$	COURSE/PROJECT: /		TEAM #:	
	TITLE:			
	PART/DWG #:		REV:	
	MATERIAL: AISI 304 Stainless	SCALE: 1:1.5	SIZE: A	SHEET: 12 OF 4

UNLESS OTHERWISE SPECIFIED DIMS ARE IN INCHES		
	DATE	INITIALS
APPROVED BY:		
INSPECTED BY:		



DRAWN BY: JOSUA A.

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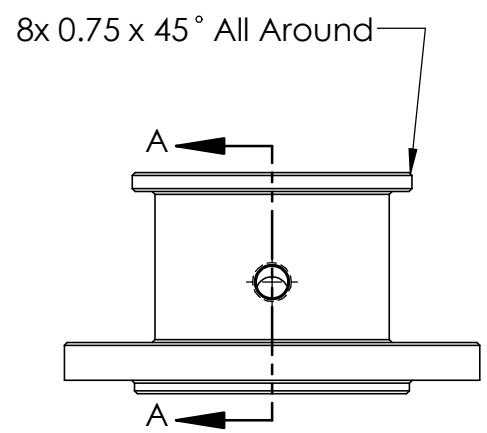
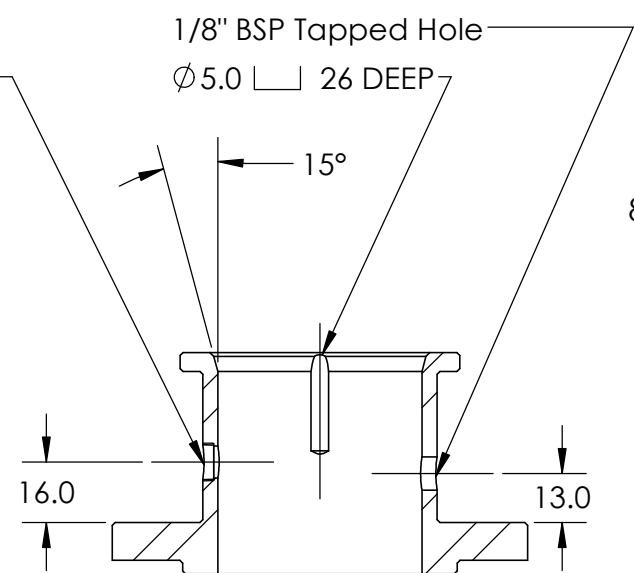
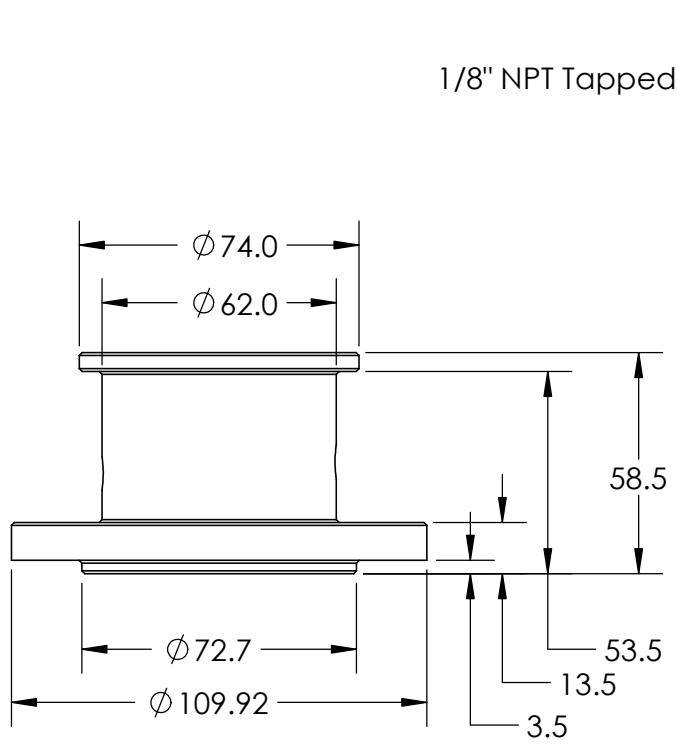
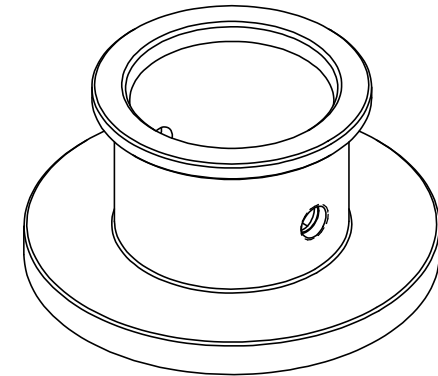
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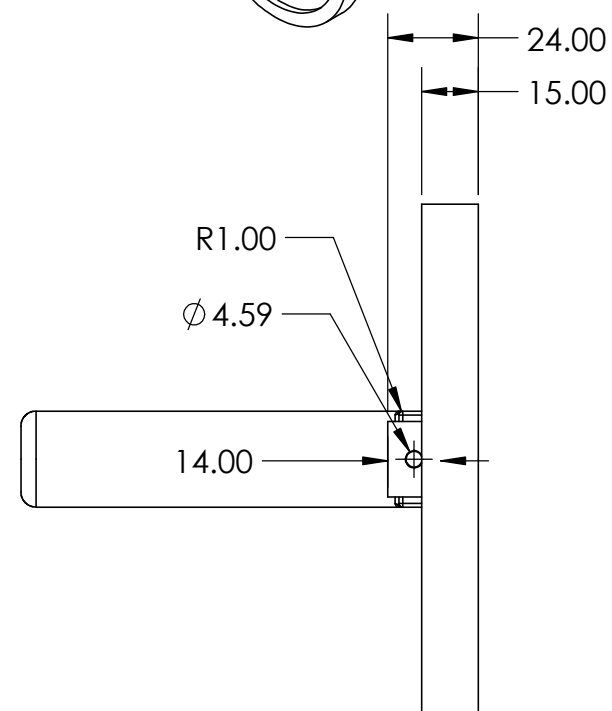
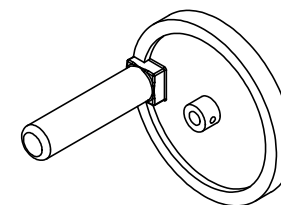
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UNLESS OTHERWISE SPECIFIED DIMS ARE IN MM			DEFAULT TOLERANCES: ANGLE ± 0.5° X ± 2.5 .X ± .25 .XX ± .13 .XXX ± .013	COURSE/PROJECT: ME 190 / Espresso Machine		TEAM #: 1	
				TITLE: Brew Chamber			
APPROVED BY: 04/26/26 GC			 SACRAMENTO STATE	PART/DWG #:		REV: 2	
				MATERIAL: AISI 316 Stainless Steel		SCALE: 1:2	SIZE: A
INSPECTED BY:				FINISH: 2B			



DEFAULT TOLERANCES: ANGLE ± 0.5° .X ± .1 .XX ± .03 .XXX ± .005	COURSE/PROJECT: ME 190 / Espresso Machine			TEAM #:
TITLE: Crank Handle/Wheel				
 SACRAMENTO STATE	PART/DWG #: 2		REV:	
	MATERIAL: PolyPropylene	SCALE: 1:5	SIZE: A	SHEET: 1 OF 1
FINISH:				

Espresso Machine - Group 1	
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Vendor Quotes are required for all items listed.

Save all packing slips and receipts for reconciliation

All purchases must be approved

Budget Edit Date 4/29/26

Item #	Category	Part	Vendor	Vendor Part #	Vendor Info/Product Link	Quote #	Unit Cost	Quantity	Cost
1	Structural	Filament 900g Polypropylene spool	Eryone (amazon)	N/A	https://a.co/d/0cQs1CHV		\$32.99	1	\$32.99
2	Structural	Filament 1kg ASA	Overture (amazon)	N/A	https://a.co/d/08ewFaC2		\$19.99	1	\$19.99
3	Structural	Latch	McMasterCarr	6082A16	https://www.mcmaster.com/6082A16/		\$9.05	1	\$9.05
4	Linkage	D-Profile Rotary Shaft	McMasterCarr	8632T38	https://www.mcmaster.com/8632T38/		\$11.83	1	\$11.83
5	Linkage	Oil-Embedded Ball Joint	McMasterCarr	6072K53	https://www.mcmaster.com/6072K53-6072K531/		\$18.32	2	\$36.64
6	Gear	Steel Worm	McMasterCarr	5754SK631	https://www.mcmaster.com/5754SK631/		\$21.79	1	\$21.79
7	Gear	Worm Gear	McMasterCarr	5754SK758	https://www.mcmaster.com/5754SK758/		\$63.68	1	\$63.68
8	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K117	https://www.mcmaster.com/6659K117/		\$2.09	2	\$4.18
9	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K224	https://www.mcmaster.com/6659K224/		\$19.43	2	\$38.86
10	Linkage	Rotary Shaft	McMasterCarr	4143N17	https://www.mcmaster.com/4143N17/		\$10.67	1	\$10.67
11	Linkage	Collar	McMasterCarr	6056N16	https://www.mcmaster.com/6056N16/		\$2.25	1	\$2.25
12	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K146	https://www.mcmaster.com/6659K146/		\$3.26	5	\$16.30
13	Linkage	Collar	McMasterCarr	6058N18	https://www.mcmaster.com/6058N18/		\$6.91	1	\$6.91
14	Structural	10mm bolt (30)	McMasterCarr	91180A634	https://www.mcmaster.com/91180A634/		\$14.90	1	\$14.90
15	Structural	10mm nut	McMasterCarr	90364A102	https://www.mcmaster.com/90364A102/		\$8.00	1	\$8.00
16	Structural	10mm bot (35)	McMasterCarr	91280A371	https://www.mcmaster.com/91280A371/		\$13.20	1	\$13.20
17	Structural	M5 Button Head Hex-Drive Screw	McMasterCarr	92095A207	https://www.mcmaster.com/92095A207/		\$17.80	1	\$17.80
18	Structural	M8 Hex Head Bolt	McMasterCarr	91280A085	https://www.mcmaster.com/91280A085/		\$10.00	1	\$10.00
19	Structural	M8 Hex Nut	McMasterCarr	94168A114	https://www.mcmaster.com/94168A114/		\$6.08	1	\$6.08
20	Structural	M4 Button Head Hex-Drive Screw	McMasterCarr	92095A476	92095a476https://www.mcmaster.com/92095A476/		\$11.55	1	\$11.55
21	Structural	304 SS BasePlate	SendCutSend	N/A	https://sendcutsend.com/		\$200.00	1	\$200.00
22	Structural	304 SS Upright	SendCutSend	N/A	https://sendcutsend.com/		\$25.00	2	\$50.00
23	Structural	304 SS ClampBracket	SendCutSend	N/A	https://sendcutsend.com/		\$40.00	1	\$40.00
24	Structural	Clamp	Westley	WB-1	https://www.linkedin.com/in/westley-baird/		\$0.00	1	\$0.00
25	Piston	Piston Pin	McMasterCarr	89535K17	https://www.mcmaster.com/89535K17-89535K171/		\$4.33	1	\$4.33
26	Piston	Piston Head	McMasterCarr	9535K41	https://www.mcmaster.com/89535K41-89535K412/		\$20.78	1	\$20.78
27	Piston	Piston Linkage	McMasterCarr	9008K12	https://www.mcmaster.com/9008K12-9008K121/		\$6.24	1	\$6.24
28	Piston	Piston O-Ring	Amazon	VUYVUY	https://www.amazon.com/VUYVUY-Ring-Seals%EF%BC%8CSilicone-Width-Gasket/dp/B0F4X4BXQH		\$24.62	1	\$24.62
29	Water	Tank Valve Gasket	In House	NA	Made in house by printing TPU			1	\$0.00
30	Water	Water Gasket	McMasterCarr	9452K114	https://www.mcmaster.com/9452K114/		\$5.43	1	\$5.43
31	Water	Tank lid Pin	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
32	Water	Tank lid	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
33	Water	Tank	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
34	Water	Tank lid Spring	McMasterCarr	9434K58	https://www.mcmaster.com/9434K58/		\$4.70	1	\$4.70
35	Heating	Resistance wire	Amazon	Ni80 wire	Ni80 wire		\$8.80	1	\$8.80
36	Heating	Copper Tubing	Harfington	a19011700ux0032	https://www.harfington.com/products/p-1058691		\$13.58	1	\$13.58
37	Pumping	Paristaltic Pump	Alibaba	NA	Pump		\$17.00	1	\$17.00
38	Brew Chamber	58mm Walnut Portafilter	Flair Espresso	608816113114	Flair 58mm Walnut Portafilter - Flair Espresso		\$72.00	1	\$72
39	Brew Chamber	Etched Puck Screen	Flair Espresso	860012366040	Etched Puck Screen for the Flair 58 Flair Espresso		\$15.00	1	\$15
40	Brew Chamber	4" x 2.5" 316 Stainless Steel Round Bar	Midwest Steel & Alum	3165SRB4	316 Stainless Steel Round Bar Midwest Steel & Aluminum		\$71.99	3	\$215.97
41	Electronics	NTC 3590B 10k ohm thermistor	Uxcell (amazon)	N/A	https://a.co/d/0h9s5I5s		\$8.29	1	\$8.29
42	Brew Chamber	Nuova Simonelli Conical head gasket	Espresso Parts	N/A	espressoparts.com		\$3.30	1	\$3.30
43	Brew Chamber	Nuova Simonelli Group Seat	Espresso Parts	N/A	espressoparts.com		\$106.24	1	\$106.24
44	Brew Chamber	Nuova Simonelli Group Head Bell O-ring	Espresso Parts	N/A	espressoparts.com		\$1.55	1	\$1.55
45	Electronics	120V AC to 24V DC 8.5A 300W PSU	Amazon	LM-HKDY-24V300W-1P	https://a.co/d/04FIDtKg		\$21.99	1	\$21.99
46	Electronics	Raspberry Pi 3B+ 1GB	Amazon	N/A	https://a.co/d/0c4nC6LW		\$53.58	1	\$50.00
47	Electronics	Omron SSR Relay AC/DC	Amazon	G3NA-225B	https://a.co/d/0hxS47YT		\$22.00	1	\$22.00
48	Electronics	Low voltage control Mosfet	Amazon	B0F6K4LX6Y	https://a.co/d/010s5cW9		\$9.88	1	\$9.88
49	Electronics	15A AC Filtered Inlet	Amazon	N/A	https://a.co/d/08MNNzCfv		\$8.59	1	\$8.59
50	Electronics	3 slot wago connector	Amazon	221-413	https://a.co/d/0eUvYTNp		\$7.95	1	\$7.95
51	Electronics	Stepper Controller	Amazon	DRV8825	https://www.amazon.com/dp/B0FF44DF9Q?ref=cm_sw_r_cp_ud_dp_ZCGG5WFBNP1WDWYM5G02_4		\$9.99	1	\$9.99
52	Electronics	Thermistor ADC module	Amazon	ADS1115	https://a.co/d/07u5bI5B		\$11.99	1	\$11.99
53	Electronics	16 AWG wire	Amazon	N/A	https://a.co/d/07uVx2WS		\$18.99	1	\$18.99
54	Brew Chamber	Water Pressure Gauge 49mm	esspressocare.com	1245067	Espresso machine Water Gauge Replacement parts for espresso Machine – Espresso Care		\$32.06	1	\$32.06
55	Electronics	18 AWG wire	Amazon	N/A	https://a.co/d/05OOJB8B		\$7.88	1	\$7.88
56	Electronics	Insulated fork U connectors	Amazon	N/A	https://a.co/d/0cvQKpmf		\$9.99	1	\$9.99
Total									\$1,335.88

Electrical Schematic

